HDB Smart Hub Interface Control Document

1 INTRODUCTION

1.1 Purpose

The purpose of this Interface Control Document (ICD) is to define the interface scope, data parameters and demarcation between the SLP/DECADA/Smart Hub and Smart Systems (SLS) to be ingested. It documents and tracks the necessary information required to effectively define the SLP/DECADA/Smart Hub system's interface as well as any rules for communicating with it.

1.2 Scope

This document covers all the interface requirements between the SLP/DECADA/Smart Hub and Smart Systems (SLS). This ICD helps to ensure compatibility between system segments and components by clearly communicating all possible inputs and outputs between the components.

1.3 Definitions and Terms

The Interface Control Document uses a few special terms like event and alarms. Events are operator actions, informational messages, tracking/auditing messages and statistical readings. When Smart Hub receives these events, it processes them based on the rules set and the event becomes an alarm if the event follows the rule.

2 SYSTEM OVERVIEW

This ICD describes the relationship between the Smart Systems (SLS) and SLP/DECADA/Smart Hub on data ingestion aspects.

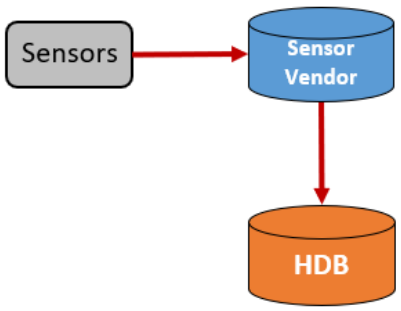
This ICD specifies the interface requirements the participating systems must meet. It describes the concept of operations for the interface, defines the message structure and protocols that govern the interchange of data, and identifies the communication paths along which the project team expects data to flow.

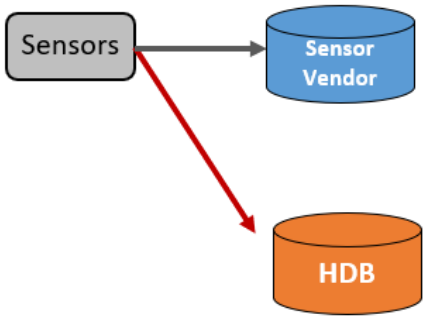
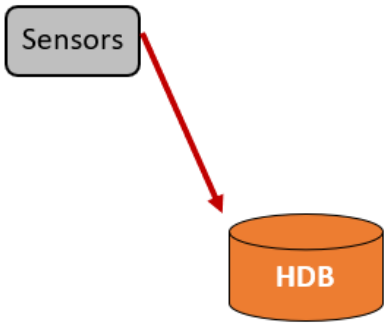
For each interface, the ICD provides the following information:

- A description of the data exchange format and protocol for exchange
- A general description of the interface
- Assumptions where appropriate
- Estimated size and frequency of data exchange
- Mandatory Security Standard

2.1 Modes of Data Transmission

Systems are to transmit data through the following three modes:

Mode	Description
1	Sensors are first connected to a vendor system/server, which would then send data to Smart Hub. 

2	Sensors send data to vendor system/servers and Smart Hub simultaneously through two separate data links. (For selected sensors only) 
3	Sensors ~~only~~ send data ~~only~~ to SLP/DECADA/Smart Hub. 

2.2 Connectivity Mode

The Contractor and the respective Specialist of the Smart System(s) shall adopt the following connectivity modes for the Systems:

	Smart System	**Mode**
1	Smart Lighting System	3

3 INTERFACE REQUIREMENTS

3.1 Interface Overview

Smart Hub provides MQTT as the interface method for data ingestion.

- Smart Hub sensor integration using MQTT protocol are divided into few steps:
  a) Framework design (Data Dictionary, Event & Alarm Design)
  b) Asset loading (Master Data)
  c) System configurations for alarms (Smart Hub admin task)
  d) Client cert generation
  e) Sending live sensor data over MQTT protocol

3.2 Protocol Allocation based on the sensor type

All Smart Systems shall use MQTT interface to transmit data to Smart Hub.

All on-site sensors for mode 1 data transmission shall also transmit in MQTT (to the vendor system/server).

3.3 Sensor Integration using MQTT protocol in detail

MQTT protocol specification details are available in

http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html

Payload format is in JavaScript Object Notation (JSON) with some mandatory key value pairs.

Sample payload:

{

"SenderId": " SmartLighting-600209-01",

"SensorId": " SmartLighting-ST-600209- 01010201",

"Resourcepath": 600209,

"EventId": "EV-600209-01010215-20160712104950272",

"EventType": "SmartLighting/motion#reading",

"Parameters": {

"sensorstatus": "online",

"Time": "2016-07-12 10:49:50.272",

"Severity": 6,

"MotionCount": 10

}

}

Highlighted in bold are mandatory fields/keys for successful communication with Smart Hub. The values for the mandatory fields/keys are explained below in detail. The non-mandatory fields/keys are designed based on the sensor type and based on how the user wants to see the information from the sensors.

3.3.1 Mandatory Field Definition for Events

Field Name	Description	Sample Value	Length in Byte
SenderId	SenderID refers to the sender of the data. Depending on the design of the Smart System, this can be the aggregation point of the data, location of the System, or the sensor itself. <SensorType>-<PostalCode>-<AggregatorIdentifier> Sensor Type must be alphabets without special characters; Instead of Postal code, location code can be used if there is no postal code available for the location. Eg. JE-OUTDOOR	Refer to table in Appendix III	255

Field Name	Description	Sample Value	Length in Byte
SensorId	<SensorType>-<Vendor>-<PostalCode>-<GenericIdentifier> Vendor name must be alphanumeric without special characters; Instead of Postal code, location code can be used if there is no postal code available for the location. E.g. JE-OUTDOOR A generic Identifier is used so that even when the specific sensor is replaced by new hardware, Smart Hub recognises it as the same sensor point.	Refer to table in Appendix III	255
Resourcepath	Resource path is the postal code of the block or the location code. This value should be same as the postal code used in the master data while loading sensor details. If postal code is not available for some sensors, then location code can be used. E.g. JE-OUTDOOR	600226	255
EventId	Reference/transaction Id. Used to group few events together. EV-<PostalCode>-<SerialNo>-<Timestamp>	EV-600216-01010215-20160712104950272	255
EventType	Format is <SensorType>/<sub-component> #<Type of event> This is further defined below.	E.g. SmartLighting/motion#reading	31
sensorstatus	To indicate sensor status List of status are online, offline, error, warning	online	10

<table>
<tr><th>Field Name</th><th>Description</th><th>Sample Value</th><th>Length in Byte</th></tr>
<tr><td>Time</td><td>yyyy-MM-dd HH:mm:ss.SSS
yyyy – Year
MM – Month
dd – Day
HH – Hour 24hr format
mm – Minute
ss – Second
SSS – Millisecond</td><td>2016-07-12 10:49:50.272</td><td>25</td></tr>
<tr><td>Severity</td><td><table>
<tr><td>1</td><td>Critical</td><td>A situation where the System is unable to operate (e.g. SmartIrrigation/moisturesensor#equipmentfault)</td></tr>
<tr><td>2</td><td>Major</td><td>Where there is a fault in some of the important equipment but it does not affect the entire operation of the System.</td></tr>
<tr><td>3</td><td>Minor</td><td>Minor equipment fault.</td></tr>
<tr><td>4</td><td>Info</td><td>It may not be an alarm but the user may be interested to know.</td></tr>
<tr><td>5</td><td>Clear</td><td>When fault is cleared.</td></tr>
<tr><td>6</td><td>Reading</td><td>While updating the system status.</td></tr>
</table>
Events such as SmartLighting/motion#reading, etc are for the updating of System status and hence the severity for such cases is ‘Reading’. For the faults, severity depends on the type of the fault.</td><td>e.g. 3</td><td>255</td></tr>
<tr><td>Parameters</td><td>All the sensor related fields are to be enclosed in the parameters. It supports only two data types -- String and Number. String values are enclosed in double quotation marks, whereas numbers are not enclosed in quotation marks.</td><td>{"LightStatus":"Normal"}</td><td>Key:255
Value:255</td></tr>
</table>

3.3.1 Mandatory Field Definition for Events (Continued)

Parameters

Parameters are the property used to indicate the dynamic values of the sensors. Since each of the sensors has a different set of functionalities, which are dynamic in nature, the current design will allow many different types of combinations of parameters to be sent to Smart Hub. Unit of measurement can also be used along with parameter names for e.g.

{“AC_voltage (V)”:230} or

{“AC_current (A)”:3.0}

Event Type

Event type is used to describe what type of event it belongs to. It is used to visualize event on the operator’s console.

The strategy for creating the event types are:

Event Type = <Type of Sensor> / <sub-type> #<event type>

E.g.: SmartLighting/motion#reading

SmartLock/Controller#intrusion

Where #reading are status updates or heart beat events (e.g. routine readings).

The frequency of the heart beat / status update is defined according to the sensor type.

3.3.2 Mandatory field definition for alarms on top of events fields

Field Name	Description	Data Type
Faultcode	Specifies type of fault as defined in the System specific dictionary	String
Description	Description or note of the fault/incident	String

Faults in the Smart Systems shall be identified and assigned severity values based on the priority of the fault. For example, if a certain event type is identified as an alarm, the alarm can be configured in the Smart Hub system configuration page (hdb->cep_config->severity.events) to get this event to appear as an alarm. Thereafter, the Smart Hub system will detect the alarms based on the severity and alarm types configured. These additional fields help to visualize the details of the alarm.

In order to clear a fault, the sensor system has to send another event to clear the previous fault generated in the same system.

Field	Description	Data Type
RefEventId	Specifies the previous fault event id	String
RefEventType	Specifies the previous fault event type	String

3.3.3 Security Requirements

MQTT Broker supports only TLS 1.2 protocol.

Supported cipher suits are:

- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,
- TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,
- TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,
- TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,
- TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384

The SSL client certificates will be issued to each of the vendor, and the sensor/system has to use this certificate to authenticate itself to the broker.

3.3.4 MQTT Structure and Protocol settings

Broker URL is:

Production: **ssl://192.168.8.35:9884**

UAT: **ssl://192.168.8.38:9884**

Topic name will be based on the sensor type.

Format for topic is **VirtualTopic/mqtt/<sensor type>/incoming**

Proposed Quality of Services setting,

Qos 2: Exactly once delivery

This is the highest quality of service, for use when neither loss nor duplication of messages are acceptable. There is an increased overhead associated with this quality of service.

A QoS 2 message has a Packet Identifier in its variable header. The receiver of a QoS 2 PUBLISH Packet acknowledges receipt with a two-step acknowledgement process.

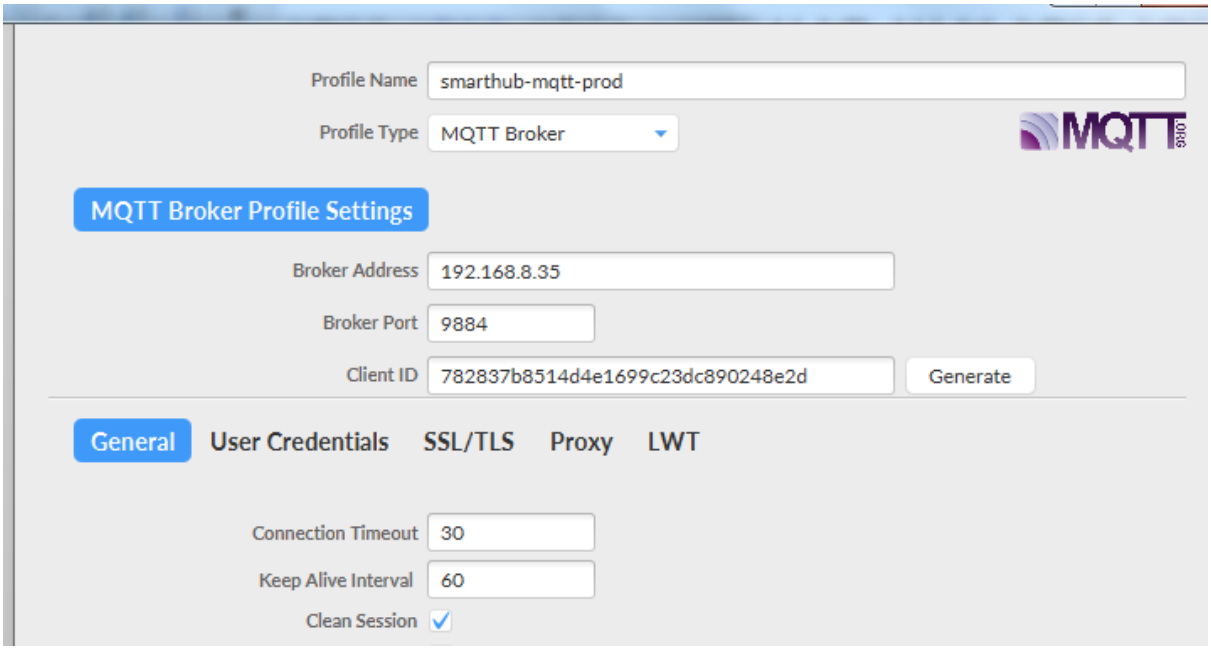
Keep live timeout for message is 60 seconds for MQTT broker to retain the established session. Heartbeat rate frequency for every sensor is by default 15 minutes, i.e. In order for the sensor to appear as online in Smart Hub, they sensor must send an event message every 15 minutes.

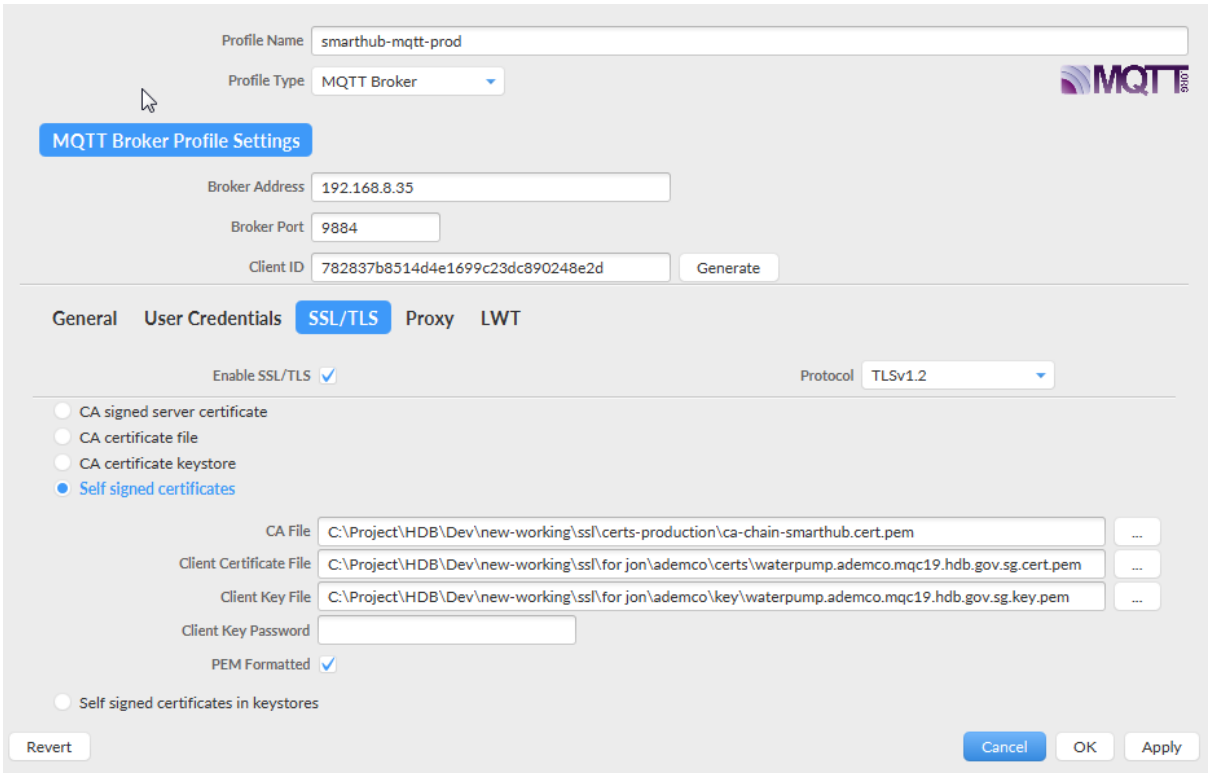
3.3.5 Testing using MQTTfx (or equivalent)

MQTTfx is an example of a free Graphical User Interface (GUI) testing tool, which helps to generate MQTT messages and send to the Smart Hub for testing.

Download MQTTfx from http://www.jensd.de/apps/mqttfx/1.7.0

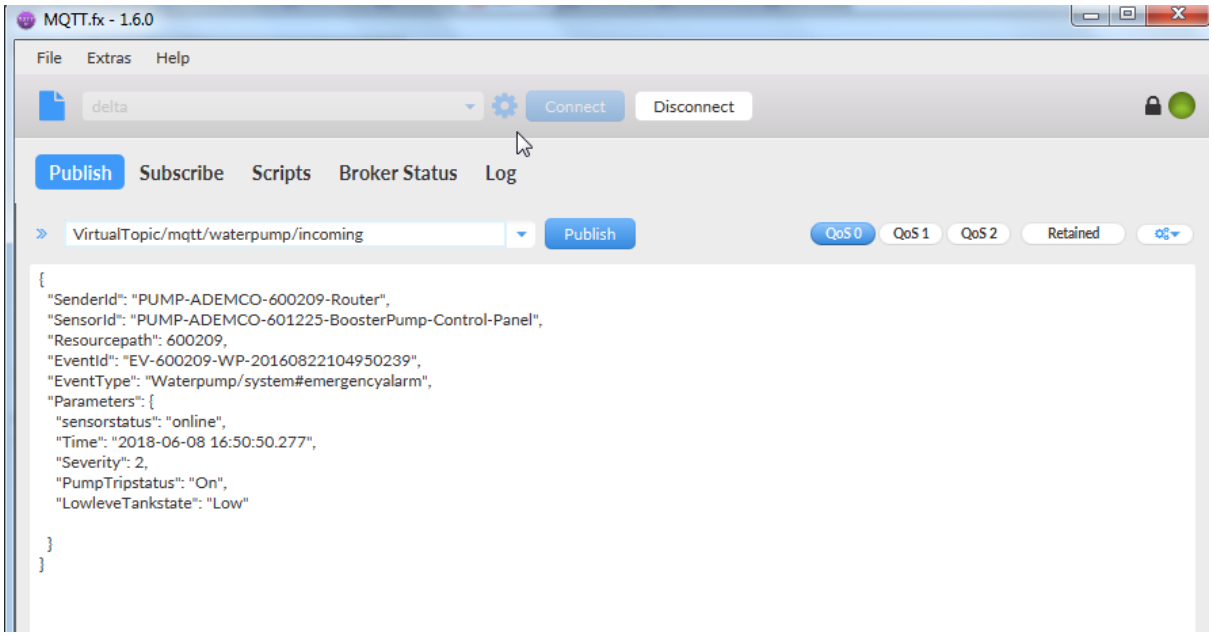
Settings needed:





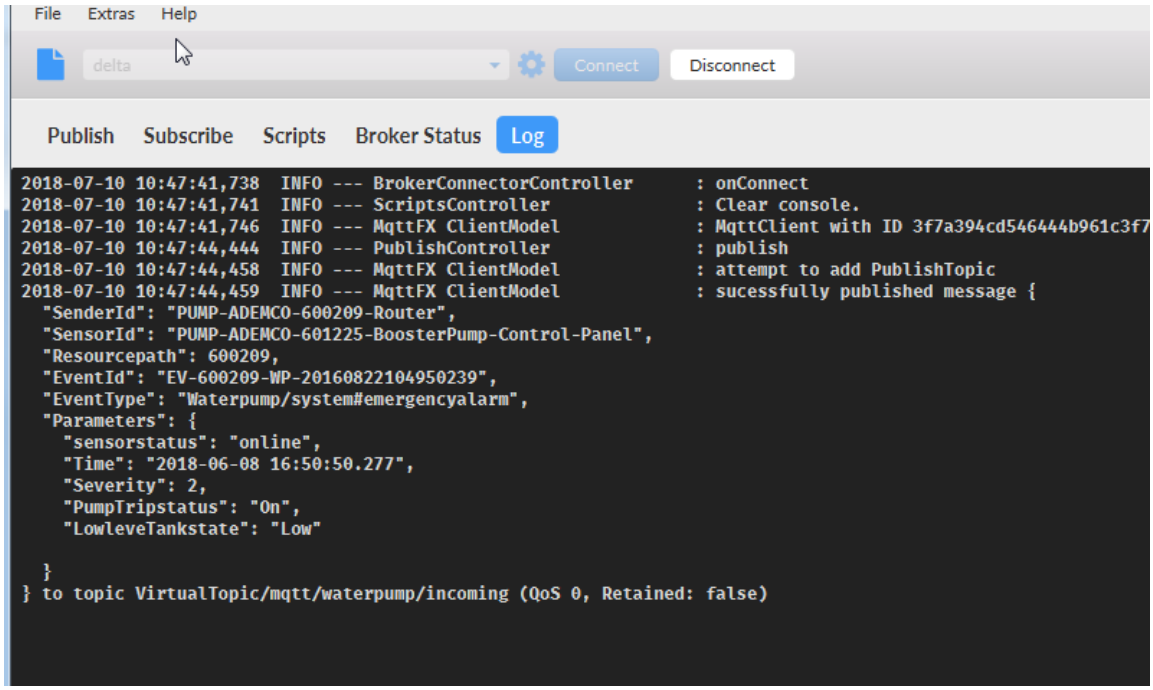
Notes:

Client Id has to be regenerated by MQTTfx;
Certificate Authority (CA) file, client certificate, client key shall be provided by HDB.



Steps for publishing message to MQTT broker:

a) Select the server you want to connect.
b) Connect and wait for green icon to appear on the right most corner
c) Create sample payload and choose topic
d) Publish the message
e) Browse logs tab for logs



4 INTERFACE DEFINITIONS IN DETAIL

The following section describes the interface definition for each sensor type in detail. In general:

The measurements readings and status changes are Events.

Faults and abnormal conditions are Alarms. The alarm is triggered based on the rules set in Smart Hub. One of the rules set is that the Smart Hub will trigger an alarm based on the severity of an event.

e.g. If severity < 4, an event turns to alarm.

4.1 Smart Lighting

4.1.1 Smart Lighting Events and Alarms

4.1.1.1 Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
Time	Time	String	25		2016-11-24 14:52:50.276	SGT
Severity	Severity	Number	8	1-6	6	
sensorstatus	Sensor status	String	20	online, offline, error	online	
GatewayStatus	Health status of gateway	String	20	Online / Offline	Online	
LightStatus	Health status of light	String	20	Normal / Dim / Flicker / Dim_Flicker Off	Normal	
SensorHealthStatus	Health status of sensor	String	20	Normal / Not_Working	Normal	
NetworkStatus	Network status of sensor	String	20	Online / Offline	Online	
ELightStatus	Health status of emergency light	String	25	Normal / Not_Responding / Emergency_Mode / Battery_Out / Battery_Low	Normal	
MotionCount	Motion detection count	Number	10		10	~~Number of motion detections per update interval~~ Motion detection count is the sum of all the bins that are with motion within the update interval.
BinStartTime	Start time of motion bins	String	25		2016-11-24 14:52:50.276	

BinCount	Number of motion bins per update interval	Number	10		90	Divide update interval into multiple bins, each bin is 10 sec
BinData	Data of motion bins per update interval	Array	20		[128, 240, 0, 0, 0, 0, 0, 0, 0, 0]	Motion bins per update interval are arranged in groups of bins in ascending order of time. Each group consists of 8 bins (one byte) and the most significant bit is the first bin and so on. Individual bin is set to 1 if motion is detected within the respective bin period.
DaylightValue	Photo sensor value	Number	10		2000	Value is in lux
TemperatureValue	Temperature sensor value	Number	10		28	Value is in Celsius
HumidityValue	Humidity sensor value	Number	10		60	Value is in relative humidity % RH
RainValue	Rain sensor value	Number	10		50	Value is in mm/Hour
EnergyUsage	Cumulative power consumption per luminaire	Number	10		1000	Value is in kWh
LightUsage	Cumulative burn time per luminaire	Number	10		100	Value is in hours
PowerValue	Power consumption under light full on condition	Number	10		300	Wh or any other value that is related to light power consumption
BlinkValue	Blink intensity under light full on condition	Number	10		100	Percent flicker, flicker index or any other value that is related to light blinking
TestTime	Date and time of emergency light test	String	25		2016-11-24 14:52:50.276	
TestType	Test type for emergency light	String	25	Functional / Duration / Commissioning	Functional	
TestResult	Test result for emergency light	String	25	Passed / Failed / Not_Tested / Test_In_Progress / Result_Not_Available	Passed	
FailureType	Type of failure for emergency light	String	25	Light_Fail / Battery_Fail / Charger_Fail /	Battery_Fail	For test failed case

				Light_Dim / Light_Flicker / Light_Dim_Flicker		
NotTested	Reason for not tested for emergency light	String	50	Not_Responding / Emergency_Mode / Test_Button_Pressed / Light_Power_Not_Calibrated / Battery_Below_Nominal_Voltage / Battery_Not_Fully_Charged / Battery_Removed / Switch_Over_Fail / Incorrect_Nominal_Battery_Voltage / Battery_Low / Charger_Fault	Battery_Removed	For not tested case
BatteryVoltage	Battery voltage after lighting test for emergency light	Number	10	3.6 to 12	3.5	Value is in voltage
RefEventType	Only for clearing early alarm	String	32			
RefEventId	Only for clearing early alarm	String	32			
Description	Description about the fault	String	512			
Faultcode	Fault code	String	32	SH-LS-OE-[106301-106400]	SH-LS-OE-106001	

4.1.1.2 Event Design

Event Type	Fields	Severity	Frequency(mins)
SmartLighting/motion#reading	sensorstatus, Time, Severity, MotionCount, BinStartTime, BinCount, BinData	6	15
SmartLighting/daylight#reading	sensorstatus, Time, Severity, DaylightValue	6	15
SmartLighting/temperature#reading	sensorstatus, Time, Severity, TemperatureValue	6	15
SmartLighting/humidity#reading	sensorstatus, Time, Severity, HumidityValue	6	15
SmartLighting/rain#reading	sensorstatus, Time, Severity, RainfallValue	6	15
SmartLighting/energyusage#reading	sensorstatus, Time, Severity, EnergyUsage	6	Once a day
SmartLighting/lightusage#reading	sensorstatus, Time, Severity, LightUsage	6	Once a day
SmartLighting/powervalue#reading	sensorstatus, Time, Severity, PowerValue, BlinkValue	6	Once a day
SmartLighting/elightresult#reading	sensorstatus, Time, Severity, TestTime, TestType, TestResult, FailureType, NotTested, BatteryVoltage	6	Ad hoc basis

4.1.1.3 Alarm Design

Alarm Type	Fields	Severity	FaultCode	Remarks
SmartLighting/gateway#fault	sensorstatus, Time, Severity, Description, Faultcode, GatewayStatus	1	SH-LS-OE-106301	
SmartLighting/light#fault	sensorstatus, Time, Severity, Description, Faultcode, LightStatus	3	SH-LS-OE-106302	
SmartLighting/sensor#fault	sensorstatus, Time, Severity, Description, Faultcode, SensorHealthStatus	3	SH-LS-OE-106303	
SmartLighting/network#fault	sensorstatus, Time, Severity, Description, Faultcode, NetworkStatus	3	SH-LS-OE-106304	
SmartLighting/eligh t#fault	sensorstatus, Time, Severity, Description, Faultcode, ELightStatus	3	SH-LS-OE-106305	
SmartLighting/system#clear	sensorstatus, Time, Severity, Description, SensorId, RefEventId, RefEventType	5		RefEventId and RefEventType should be the same values as fault EventId and EventType respectively for clearing

4.1.2 Smart Lighting Configuration and Control

4.1.2.1 Interface Overview

Smart Lighting System provides MQTT as the interface method for configuration and control. Integration between Smart Lighting System and HDB Backend System (SLP) using MQTT protocol are divided into few steps:

a) Framework design (Data Dictionary, Configuration Design)

b) Client cert generation

c) Sending set and get command over MQTT protocol

4.1.2.2 System Integration using MQTT Protocol

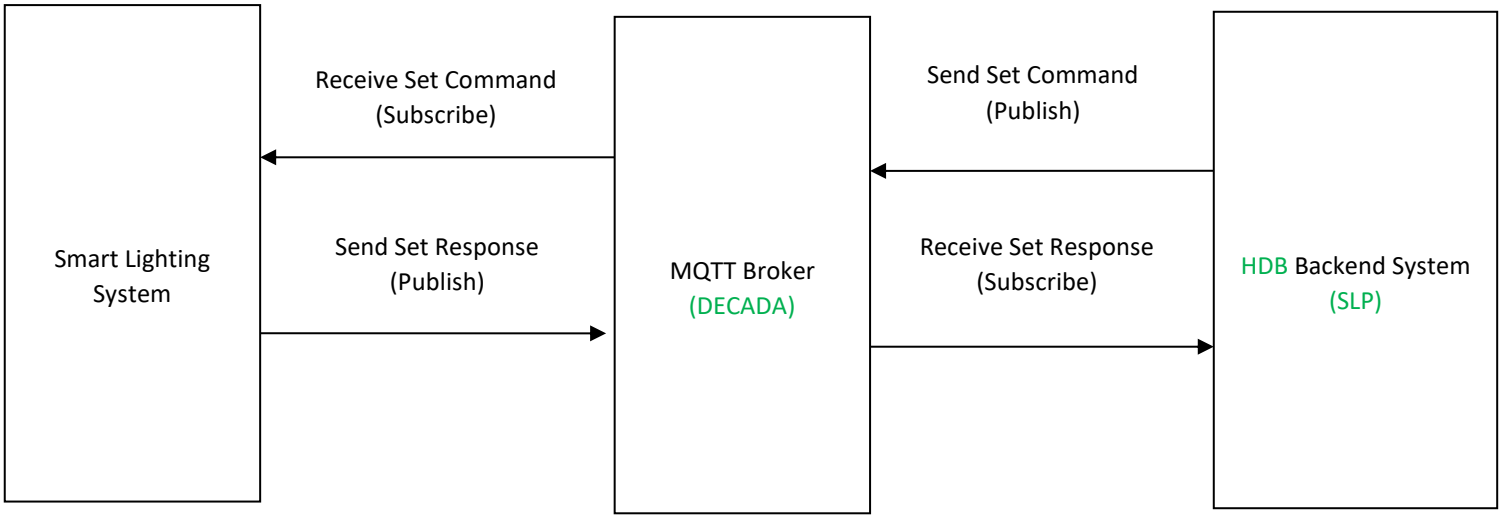
MQTT protocol specification details are available in

http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html

Payload format is json with some mandatory key value pairs.

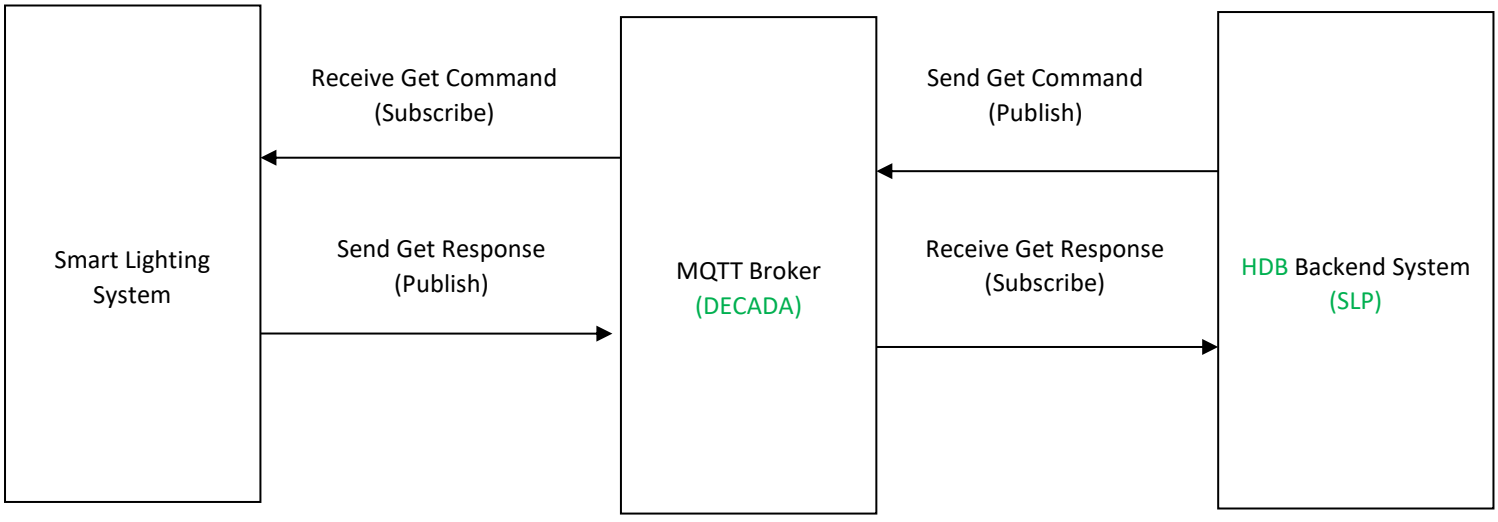
Set Command

Set command is used by HDB Backend System (SLP) to control or apply settings to a sensor or a group of sensors.



Get Command

Get command is used by HDB Backend System (SLP) to get settings from a sensor.



Get Command (Unicast)

Get command is used to retrieve the settings of a sensor based on sensor ID. It is a unicast command.

Sample payload for GET command:

```
{
   "SenderId": "Backend-600123-00",
   "SensorId": "SmartLighting-ST-600209-01010215",
   "Resourcepath": 600209,
   "EventId": "EV-600209-01010215-20160822104950234",
   "EventType": "SmartLighting/getLightOverride",
   "Parameters": {
        “CommandType”:”Get”
   }
}
```

Sample payload for GET command response:

```
{
   "SenderId": "SmartLighting-600209-01",
   "SensorId": "SmartLighting-ST-600209-01010215",
   "Resourcepath": 600209,
   "EventId": "EV-600209-01010215-20160822104950234",
   "EventType": "SmartLighting/getLightOverride",
   "Parameters": {
        “CommandType”:”Get_Ack”,
        “LightingControl”:”Auto”
   }
}
```

Set Command (Unicast)

Set command with unicast transmission is used to configure a single sensor based on sensor ID. In this case, configuration zone is not used and shall be set to 0.

Sample payload for unicast SET command:

{
"SenderId": "Backend-600123-00",
"SensorId": "SmartLighting-ST-600209-01010215",
"Resourcepath": 600209,
"EventId": "EV-600209-01010215-20160822104950234",
"EventType": "SmartLighting/setLightOverride",
"Parameters": {
“CommandType”:”Set”,
“Transmission”:”Unicast”,
“ConfigurationZone”:0,
“LightingControl”:”Auto”
}
}

Sample payload for unicast SET command response:

{
"SenderId": "SmartLighting-600209-01",
"SensorId": "SmartLighting-ST-600209-01010215",
"Resourcepath": 600209,
"EventId": "EV-600209-01010215-20160822104950234",
"EventType": "SmartLighting/setLightOverride",
"Parameters": {
“CommandType”:”Set_Ack”,
“Transmission”:”Unicast”,
“ConfigurationZone”:0,
“LightingControl”:”Auto”
}
}

Set Command (Multicast)

Set command with multicast transmission is used to configure a group of sensors under a gateway based on gateway id (sensor id field) and zone id.

Sample payload for multicast SET command:

{

"SenderId": "Backend-600123-00",

"SensorId": "SmartLighting-ST-600209-01000000",

"Resourcepath": 600209,

"EventId": "EV-600209-01000000-20160822104950234",

"EventType": "SmartLighting/setLightOverride",

"Parameters": {

“CommandType”:”Set”,

“Transmission”:”Multicast”,

“ConfigurationZone”:1,

“LightingControl”:”Auto”

}

}

Sample payload for multicast SET command response:

{

"SenderId": "SmartLighting-600209-01",

"SensorId": "SmartLighting-ST-600209-01000000",

"Resourcepath": 600209,

"EventId": "EV-600209-01000000-20160822104950234",

"EventType": "SmartLighting/setLightOverride",

"Parameters": {

“CommandType”:”Set_Ack”,

“Transmission”:”Multicast”,

“ConfigurationZone”:1,

“LightingControl”:”Auto”

}

}

Set Command (Broadcast)

Set command with broadcast transmission is used to configure all sensors under a gateway based on gateway id (sensor id field). In this case, configuration zone is not used and shall be set to 0.

Sample payload for broadcast SET command:


```
{
   "SenderId": "Backend-600123-00",
   "SensorId": "SmartLighting-ST-600209-01000000",
   "Resourcepath": 600209,
   "EventId": "EV-600209-01000000-20160822104950234",
   "EventType": "SmartLighting/setLightOverride",
   "Parameters": {
        “CommandType”:”Set”,
        “Transmission”:”Broadcast”,
        “ConfigurationZone”:0,
        “LightingControl”:”Auto”
   }
}
```


Sample payload for broadcast SET command response:


```
{
   "SenderId": "SmartLighting-600209-01",
   "SensorId": "SmartLighting-ST-600209-01000000",
   "Resourcepath": 600209,
   "EventId": "EV-600209-01000000-20160822104950234",
   "EventType": "SmartLighting/setLightOverride",
   "Parameters": {
        “CommandType”:”Set_Ack”,
        “Transmission”:”Broadcast”,
        “ConfigurationZone”:0,
```

```
        "LightingControl":"Auto"
    }
}
```

Highlighted in bold are mandatory fields/keys for successful communication with HDB Backend System (SLP). The values for the mandatory fields/keys are explained below in detail. The non-mandatory fields/keys are designed based on the parameter type.

4.1.2.2.1 Mandatory Field Definition for Events(Smart Lighting)

Field Name	Description	Sample Value	Length in Byte
SenderId	SenderID refers to the sender of the data. Depending on the design of the Smart System, this could be the aggregation point of the data, location of the System, or the sensor itself. <SensorType>-<PostalCode>-<AggregatorIdentifier> System Type must be alphabets without special characters; Instead of Postal code, location code can be used if there is no postal code available for the location. Eg. JE-OUTDOOR Note: Set <SystemType> to "Backend" for command initiated by HDB Backend System (SLP).	Refer to table in Appendix III	255
SensorId	<SensorType>-<Vendor>-<PostalCode>-<GenericIdentifier> Vendor name must be alphanumeric without special characters; Instead of Postal code, location code can be used if there is no postal code available for the location. E.g. JE-OUTDOOR A generic Identifier is used so that even when the specific sensor is replaced by new hardware, Smart Hub recognises it as the same sensor point.	Refer to table in Appendix III	255

Field Name	Description	Sample Value	Length in Byte
Resourcepath	Resource path is the postal code of the block or the location code. This value should be same as the postal code used in the Master Data while loading sensor details. If postal code is not available, then location code can be used. E.g. JE-OUTDOOR	600226	255
EventId	Reference/transaction Id. Used to group few events together. EV-<PostalCode>-< GenericIdentifier>-<Timestamp>	EV-600216-01010215-20160712104950272	255
EventType	Format is <SystemType>/<parameter type>	E.g. SmartLighting/setLightingOverride	31
CommandType	Indicating set / get request and response: - Set - Set_Ack - Set_Error - Get - Get_Ack - Get_Error Note: Error of set / get command can be due to sensor offline or incorrect parameters.	Set	20
Transmission	Indicating transmission type for set command: - Unicast - Multicast - Broadcast Note: - Unicast is for single sensor (select by sensor id). - Multicast is for a group of sensors under a gateway (select by gateway id and configuration zone). - Broadcast is for all sensors under a gateway (select by gateway id).	Unicast	20

Field Name	Description	Sample Value	Length in Byte
ConfigurationZone	Indicating lighting zone (1 to 63) to be configured and is meant for multicast command only. For unicast and broadcast command, configuration zone is set to 0.	1	10
Parameters	All the sensor related fields are to be enclosed in the parameters. It supports only two data types -- String and Number. String values are enclosed in double quotation marks, whereas numbers are not enclosed in quotation marks.	{"LightingControl":"Auto"}	Key:255 Value:255

Parameters

Parameters are the property used to indicate the dynamic values of the sensors. Since each of the sensors has a different set of functionalities, which are dynamic in nature, the current design will allow many different types of combinations of parameters.

Event Type

Event type is used to describe what type of event it belongs to. It is used to visualize event on the operator's console.

The strategy for creating the event types are:

Event Type = <Type of System>/<parameter type>

Example:

SmartLighting/setLightOverride

4.1.2.2.2 MQTT Structure and Protocol settings

Topic to Publish and Subscribe

HDB Backend System Command:

HDB Backend System (SLP) publishes to the following topic to send command to a specific smart lighting system based on postal code. The command is defined in the event type.

VirtualTopic/mqtt/SmartLighting/<postal code>

HDB Backend System (SLP) subscribes to the following topic to receive command response from smart lighting system. The command response is defined in the event type.

VirtualTopic/mqtt/SmartLighting/response

Smart Lighting System Response:

Smart Lighting System subscribes to the following topic to receive command from HDB Backend System (SLP). The command is defined in the event type.

VirtualTopic/mqtt/SmartLighting/<postal code>

Smart Lighting System publishes to the following topic to send command response to HDB Backend System (SLP). The command response is defined in the event type.

VirtualTopic/mqtt/SmartLighting/response

Proposed Quality of Services setting,

Qos 2: Exactly once delivery

This is the highest quality of service, for use when neither loss nor duplication of messages are acceptable. There is an increased overhead associated with this quality of service.

A QoS 2 message has a Packet Identifier in its variable header. The receiver of a QoS 2 PUBLISH Packet acknowledges receipt with a two-step acknowledgement process.

Security Requirements:

MQTT Broker supports only TLS 1.2 protocol.

4.1.2.3 Interface Definitions in Detail

4.1.2.3.1 Lighting Zones

This command is used to assign up to 8 lighting zones to a sensor. Lighting zone consists of a group of sensors and is typically used for device configuration and lighting control (e.g. predictive lighting, dimming profile, etc). Total 63 lighting zones are supported per gateway.

Event Type	Command Type	Fields	Remarks
SmartLighting/setLightingZone	Set	Transmission, ConfigurationZone, NumberOfZone, ZoneId#1 … ZoneId#n	Assign lighting zones to a sensor (unicast command)
	Set_Ack	Transmission, ConfigurationZone, NumberOfZone, ZoneId#1 … ZoneId#n	
SmartLighting/getLightingZone	Get		Get settings of lighting zones from a sensor
	Get_Ack	NumberOfZone, ZoneId#1 … ZoneId#n	

Sample parameters for SET command (SmartLighting/setLightingZone)

"Parameters": {

“CommandType”:”Set”,

“Transmission”:”Unicast”,

“ConfigurationZone”:0,

“NumberOfZone”:2,

“LightingZones”:[

{

“ZoneId”:10,

},

{

“ZoneId”:15,

}

]

}

Sample parameters for GET command response (SmartLighting/getLightingZone)

"Parameters": {

“CommandType”:”Get_Ack”,

```
"NumberOfZone":2,
"LightingZones":[
        {
                "ZoneId":10,
        },
        {
                "ZoneId":15,
        }
    ]
}
```

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
NumberOfZone	Number of lighting zones assigned to a sensor	Number	10	1 to 8	5	
ZoneId	Lighting zone id	Number	10	1 to 63	1	

4.1.2.3.2 Trigger Zones

This command is used to assign up to 8 trigger zones to a sensor. In this case, the sensor is considered as a trigger node and the trigger zones are actually lighting zones being activated by the trigger node. When the sensor detects movement, the lights in the corresponding trigger zones (or lighting zones) will be turned on. Trigger zones are typically used by predictive lighting to turn on lights in multiple paths.

Event Type	Command Type	Fields	Remarks
SmartLighting/setTriggerZone	Set	Transmission, ConfigurationZone, NumberOfZone, ZoneId#1 … ZoneId#n	Assign trigger zones to a sensor (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, NumberOfZone, ZoneId#1 … ZoneId#n	
SmartLighting/getTriggerZone	Get		Get settings of trigger zones from a sensor

	Get_Ack	NumberOfZone, ZoneId#1 … ZoneId#n	

Sample parameters for SET command (SmartLighting/setTriggerZone)

```
"Parameters": {
    “CommandType”:”Set”,
    “Transmission”:”Unicast”,
    “ConfigurationZone”:0,
    “NumberOfZone”:2,
    “TriggerZones”:[
        {
            “ZoneId”:10,
        },
        {
            “ZoneId”:15,
        }
    ]
}
```

Sample parameters for GET command response (SmartLighting/getTriggerZone)

```
"Parameters": {
    “CommandType”:”Get_Ack”,
    “NumberOfZone”:2,
    “TriggerZones”:[
        {
            “ZoneId”:10,
        },
        {
            “ZoneId”:15,
        }
```

]

}

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
NumberOfZone	Number of trigger zones assigned to a sensor	Number	10	1 to 8	5	
ZoneId	Trigger zone id	Number	10	1 to 63	1	

4.1.2.3.3 Lighting Configuration

The following commands are used to configure lighting parameters.

Event Type	Command Type	Fields	Remarks
SmartLighting/setLightOverride	Set	Transmission, ConfigurationZone, LightingControl, FailsafeDelay	Override light to auto, force on or force off mode
	Set_Ack	Transmission, ConfigurationZone, LightingControl, FailsafeDelay	
SmartLighting/getLightOverride	Get		Get light override mode
	Get_Ack	LightingControl, FailsafeDelay	
SmartLighting/setLightIntensity	Set	Transmission, ConfigurationZone, DimIntensity, BrightIntensity	Set dim and bright intensity
	Set_Ack	Transmission, ConfigurationZone, DimIntensity, BrightIntensity	
SmartLighting/getLightIntensity	Get		Get dim and bright intensity
	Get_Ack	DimIntensity, BrightIntensity	
SmartLighting/setSoftDimming	Set	Transmission, ConfigurationZone, RampUpTime, RampDownTime	Set ramp up and ramp down time
	Set_Ack	Transmission, ConfigurationZone, RampUpTime, RampDownTime	
SmartLighting/getSoftDimming	Get		Get ramp up and ramp down time
	Get_Ack	RampUpTime, RampDownTime	

SmartLighting/setTimeDelay	Set	Transmission, ConfigurationZone, HoldTime	Set light hold time
	Set_Ack	Transmission, ConfigurationZone, HoldTime	
SmartLighting/getTimeDelay	Get		Get light hold time
	Get_Ack	HoldTime	
SmartLighting/setLightInstallDate	Set	Transmission, ConfigurationZone, InstallDate	Set light installed date Only applicable to unicast and broadcast command (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, InstallDate	
SmartLighting/getLightInstallDate	Get		Get light installed date
	Get_Ack	InstallDate	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
LightingControl	Set lighting operation mode	String	20	Auto / Force_On / Force_Off ~~Unoccupy_Off~~	Auto	Auto: On-demand lighting based on motion detection. Force_On: Light full on. Force_Off: Light full off. ~~Unoccupy_Off: Turn off light in unoccupy area~~
FailsafeDelay	Delay to enter failsafe mode upon lost sync	Number	10	0 to 250 min	5	Set to 0 to enter failsafe mode immediately upon lost sync
DimIntensity	Dim intensity of light	Number	10	0 to 50	20	Value is in percentage
BrightIntensity	Bright intensity of light	Number	10	50 to 100	80	Value is in percentage
RampUpTime	Ramp up time from dim to bright	Number	10	1 to 3	1	Value is in second
RampDownTime	Ramp down time from bright to dim	Number	10	1 to 3	2	Value is in second
HoldTime	Time interval to keep light on upon motion detection	Number	10	10 to 3600	30	Value is in second

InstallDate	Light installed date	String	20	dd-MM-yyyy dd – Day MM – Month yyyy – Year	01-10-2020	

4.1.2.3.4 Sensor Configuration

The following commands are used to configure sensor parameters.

Event Type	Command Type	Fields	Remarks
SmartLighting/setMotionSensitivity	Set	Transmission, ConfigurationZone, MotionSensitivity	Set motion detection range
	Set_Ack	Transmission, ConfigurationZone, MotionSensitivity	
SmartLighting/getMotionSensitivity	Get		Get motion detection range
	Get_Ack	MotionSensitivity	
SmartLighting/setPhotoSensing	Set	Transmission, ConfigurationZone, PhotoSensing, PhotoUpperThreshold, PhotoLowerThreshold	Set threshold values of photo sensor
	Set_Ack	Transmission, ConfigurationZone, PhotoSensing, PhotoUpperThreshold, PhotoLowerThreshold	
SmartLighting/getPhotoSensing	Get		Get threshold values of photo sensor
	Get_Ack	PhotoSensing, PhotoUpperThreshold, PhotoLowerThreshold	
SmartLighting/setRainSensing	Set	Transmission, ConfigurationZone, RainSensing, RainUpperThreshold, RainLowerThreshold	Set threshold values of rain sensor
	Set_Ack	Transmission, ConfigurationZone, RainSensing, RainUpperThreshold, RainLowerThreshold	
SmartLighting/getRainSensing	Get		Get threshold values of rain sensor
	Get_Ack	RainSensing, RainUpperThreshold,	

		RainLowerThreshold	

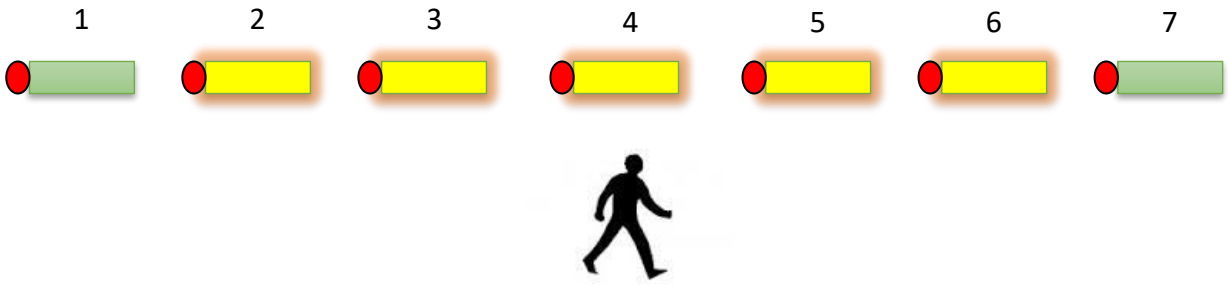
Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
MotionSensitivity	3 levels of motion detection range	Number	10	1 to 3	2	Level 1 to 3 is corresponding to low, mid and high detection range which may vary according to different sensor types (e.g. indoor / outdoor sensor).
PhotoSensing	Photo sensing control	String	20	Enable / Disable	Enable	
PhotoUpperThreshold	Upper threshold of photo sensor to detect bright daylight	Number	10	50 to 1000	400	Value is in lux
PhotoLowerThreshold	Lower threshold of photo sensor to detect dim daylight	Number	10	50 to 1000	200	Value is in lux
RainSensing	Rain sensing control	String	20	Enable / Disable	Enable	
RainUpperThreshold	Upper threshold of rain sensor to detect heavy rain	Number	10	5 to 255	30	Value is in terms of number of pulses per minute (each pulse is 0.01mm of rainfall)
RainLowerThreshold	Lower threshold of rain sensor to detect light rain	Number	10	5 to 255	5	Value is in terms of number of pulses per minute (each pulse is 0.01mm of rainfall)

4.1.2.3.5 Proximity Lighting

Proximity lighting is used to turn on lights in advance in a straight path where the pedestrian is heading. This is achieved by listening to the neighbour nodes and turning on light whenever neighbour nodes detect motion.

For example, assuming there are 7 sensors in a row which are configured to listen to 2 immediate neighbours. If motion is detected by say sensor 4, which will turn on its own light and lights in sensor 2, 3, 5, and 6 will be tuned on as well by the respective sensors.



Event Type	Command Type	Fields	Remarks
SmartLighting/setProximityLighting	Set	Transmission, ConfigurationZone, LightingControl, SaveNodeSetting, NumberOfNeighbourNode	Configure proximity lighting
	Set_Ack	Transmission, ConfigurationZone, LightingControl, SaveNodeSetting, NumberOfNeighbourNode	
SmartLighting/getProximityLighting	Get		Get settings of proximity lighting
	Get_Ack	LightingControl, NumberOfNeighbourNode	

Data Dictionary

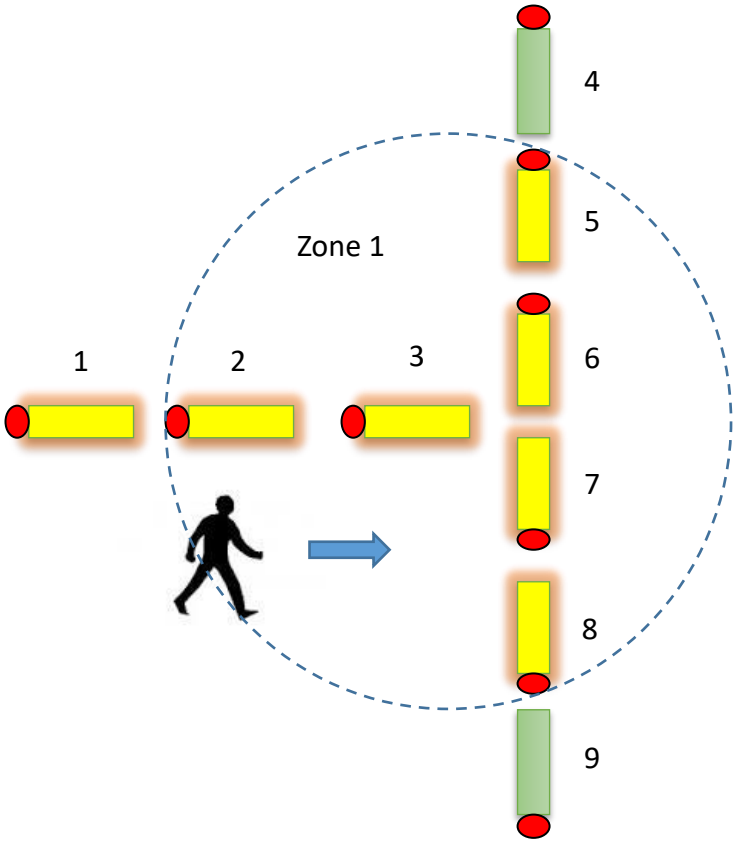
Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
LightingControl	Proximity lighting control	String	20	Enable / Disable	Enable	
SaveNodeSetting	Indicating whether to save the setting of number of neighbour nodes	String	20	Yes / No	Yes	
NumberOfNeighbourNode	Listen to number of immediate neighbour nodes for motion status	Number	10	1 to 3	3	

4.1.2.3.6 Predictive Lighting

Predictive lighting is used to turn on lights in open areas and multiple paths such that when motion is detected by a sensor (trigger node), the lights in the corresponding trigger zones will be turned on.

For example, assuming there are 9 sensors deployed at the T-junction. Sensor 2, 3, 5, 6, 7 and 8 are configured as lighting zone 1 and these sensors are also configured

as the trigger nodes that have zone 1 as the trigger zone. In this case, any motion detected by these sensors will turn on the lights in zone 1.



Event Type	Command Type	Fields	Remarks
SmartLighting/setPredictiveLighting	Set	Transmission, ConfigurationZone, LightingControl	Configure predictive lighting (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, LightingControl	
SmartLighting/getPredictiveLighting	Get		Get settings of predictive lighting (unicast command to gateway)
	Get_Ack	LightingControl	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
LightingControl	Predictive lighting control	String	20	Enable / Disable	Enable	

4.1.2.3.7 Timer Profiles

Timer profiles are used to control lighting operation mode based on the time of day, a total 10 profiles are supported. For example, it can be used to turn off lights from 7am to 7pm.

Event Type	Command Type	Fields	Remarks
SmartLighting/setTimerProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, Action [Profile 2] ZoneId, StartTime, EndTime, DayOfWeek, Action … [Profile n] ZoneId, StartTime, EndTime, DayOfWeek, Action	Configure up to 10 timer profiles (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, Action [Profile 2] ZoneId, StartTime, EndTime, DayOfWeek, Action … [Profile n] ZoneId, StartTime, EndTime, DayOfWeek, Action	
SmartLighting/getTimerProfile	Get		Get settings of timer profiles (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatAction,	

		NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, Action [Profile 2] ZoneId, StartTime, EndTime, DayOfWeek, Action … [Profile n] ZoneId, StartTime, EndTime, DayOfWeek, Action	

Note: [Profile #] in the above table is not the actual field, it is only meant for illustration purpose.

Sample parameters for SET command (SmartLighting/setTimerProfile)

```
"Parameters": {
    “CommandType”:”Set”,
    “Transmission”:”Unicast”,
    “ConfigurationZone”:0,
    “ProfileControl”:”Enable”,
    “RepeatAction”:30,
    “NumberOfProfile”:2,
    “TimerProfiles”: [
        {
            “ZoneId”:1,
            “StartTime”:”07:00”,
            “EndTime”:”19:00”,
            “DayOfWeek”:”Everyday”,
            “Action”:”Force_Off”
        },
        {
            “ZoneId”:1,
```

```
                “StartTime”:”19:00”,
                “EndTime”:”07:00”,
                “DayOfWeek”:”Everyday”,
                “Action”:”Auto”
            }
        ]
    }
```

Sample parameters for GET command response (SmartLighting/getTimerProfile)

```
    "Parameters": {
        “CommandType”:”Get_Ack”,
        “ProfileControl”:”Enable”,
        “RepeatAction”:30,
        “NumberOfProfile”:2,
        “TimerProfiles”: [
            {
                “ZoneId”:1,
                “StartTime”:”07:00”,
                “EndTime”:”19:00”,
                “DayOfWeek”:”Everyday”,
                “Action”:”Force_Off”
            },
            {
                “ZoneId”:1,
                “StartTime”:”19:00”,
                “EndTime”:”07:00”,
                “DayOfWeek”:”Everyday”,
                “Action”:”Auto”
            }
        ]
```

}

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatAction	Repeat action command per interval	Number	10	0 to 250 min	30	Set to 0 to send action command once.
NumberOfProfile	Number of profiles to be set	Number	10	1 to 10	5	
ZoneId	Lighting zone to apply setting when event is occurred	Number	10	1 to 63 / 255	255	Set to 255 to select all lights under a gateway
StartTime	Start time of event	String	20	HH:mm HH – Hour mm – Minute	07:00	
EndTime	End time of event	String	20	HH:mm HH – Hour mm – Minute	19:00	
DayOfWeek	Select day of week to monitor event	String	20	Weekday / Weekend / Everyday / Holiday	Everyday	
Action	Perform lighting control when event is occurred	String	20	Auto / Force_On / Force_Off ~~Unoccupy_Off~~	Auto	

4.1.2.3.8 Dimming Profiles

Dimming profiles are used to control dim and bright intensity based on time of day, total 10 profiles are supported. For example, it can be used to set the dim intensity of lights to a lower level after midnight to further save energy.

Event Type	Command Type	Fields	Remarks
SmartLighting/setDimmingProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel [Profile 2] ZoneId, StartTime, EndTime,	Configure up to 10 dimming profiles (unicast command to gateway)

		DayOfWeek, DimLevel, BrightLevel … [Profile n] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel	
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel [Profile 2] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel … [Profile n] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel	
SmartLighting/getDimmingProfile	Get		Get settings of dimming profiles (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel [Profile 2] ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel … [Profile n]	

		ZoneId, StartTime, EndTime, DayOfWeek, DimLevel, BrightLevel	

Note: [Profile #] in the above table is not the actual field, it is only meant for illustration purpose.

Sample parameters for SET command (SmartLighting/setDimmingProfile)

```
"Parameters": {
    “CommandType”:”Set”,
    “Transmission”:”Unicast”,
    “ConfigurationZone”:0,
    “ProfileControl”:”Enable”,
    “RepeatAction”:30,
    “NumberOfProfile”:2,
    “DimmingProfiles”: [
        {
            “ZoneId”:1,
            “StartTime”:”07:00”,
            “EndTime”:”23:59”,
            “DayOfWeek”:”Everyday”,
            “DimLevel”:20,
            “BrightLevel”:254
        },
        {
            “ZoneId”:1,
            “StartTime”:”00:00”,
            “EndTime”:”07:00”,
            “DayOfWeek”:”Everyday”,
            “DimLevel”:10
            “BrightLevel”:254
```

```
            }
        ]
    }
```

Sample parameters for GET command response (SmartLighting/getDimmingProfile)

```
    "Parameters": {
        “CommandType”:”Get_Ack”,
        “ProfileControl”:”Enable”,
        “RepeatAction”:30,
        “NumberOfProfile”:2,
        “DimmingProfiles”: [
            {
                “ZoneId”:1,
                “StartTime”:”07:00”,
                “EndTime”:”23:59”,
                “DayOfWeek”:”Everyday”,
                “DimLevel”:20
                “BrightLevel”:254
            },
            {
                “ZoneId”:1,
                “StartTime”:”00:00”,
                “EndTime”:”07:00”,
                “DayOfWeek”:”Everyday”,
                “DimLevel”:10
                “BrightLevel”:254
            }
        ]
    }
```

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatAction	Repeat action command per interval	Number	10	0 to 250 min	30	Set to 0 to send action command once.
NumberOfProfile	Number of profiles to be set	Number	10	1 to 10	5	
ZoneId	Lighting zone to apply setting when event is occurred	Number	10	1 to 63 / 255	255	Set to 255 to select all lights under a gateway
StartTime	Start time of event	String	20	HH:mm HH – Hour mm – Minute	07:00	
EndTime	End time of event	String	20	HH:mm HH – Hour mm – Minute	19:00	
DayOfWeek	Select day of week to monitor event	String	20	Weekday / Weekend / Everyday / Holiday	Everyday	
DimLevel	Set dim intensity when event is occurred	Number	10	0 to 50 / 254 / 255	20	Value is in percentage Set to 254 to keep current dim level unchanged Set to 255 to use default dim level
BrightLevel	Set bright intensity when event is occurred	Number	10	50 to 100 / 254 / 255	80	Value is in percentage Set to 254 to keep current bright level unchanged Set to 255 to use default bright level

4.1.2.3.9 Luminance Profiles

Luminance profiles are used to set the luminance level of lights based on predefined schedule, total 10 profiles are supported. For example, it can be used to set the luminance level of new lights to a lower level initially and subsequently increase the luminance level over the years to compensate LED decay.

Event Type	Command Type	Fields	Remarks
SmartLighting/setLuminanceProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartDate, EndDate, LuminanceLevel [Profile 2] ZoneId, StartDate, EndDate, LuminanceLevel … [Profile n] ZoneId, StartDate, EndDate, LuminanceLevel	Configure up to 10 luminance profiles (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartDate, EndDate, LuminanceLevel [Profile 2] ZoneId, StartDate, EndDate, LuminanceLevel … [Profile n] ZoneId, StartDate, EndDate, LuminanceLevel	
SmartLighting/getLuminanceProfile	Get		Get settings of luminance profiles (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatAction, NumberOfProfile, [Profile 1] ZoneId, StartDate, EndDate, LuminanceLevel [Profile 2] ZoneId, StartDate, EndDate, LuminanceLevel …	

		[Profile n] ZoneId, StartDate, EndDate, LuminanceLevel	

Note: [Profile #] in the above table is not the actual field, it is only meant for illustration purpose.

Sample parameters for SET command (SmartLighting/setLuminanceProfile)

```
"Parameters": {
    “CommandType”:”Set”,
    “Transmission”:”Unicast”,
    “ConfigurationZone”:0,
    “ProfileControl”:”Enable”,
    “RepeatAction”:1,
    “NumberOfProfile”:2,
    “LuminanceProfiles”: [
        {
            “ZoneId”:1,
            “StartDate”:”01-01-2020”,
            “EndDate”:”31-12-2020”,
            “LuminanceLevel”:80
        },
        {
            “ZoneId”:1,
            “StartDate”:”01-01-2021”,
            “EndDate”:”31-12-2021”,
            “LuminanceLevel”:85
        }
    ]
}
```

Sample parameters for GET command response (SmartLighting/getLuminanceProfile)

"Parameters": {

“CommandType”:”Get_Ack”,

“ProfileControl”:”Enable”,

“RepeatAction”:1,

“NumberOfProfile”:2,

“LuminanceProfiles”: [

{

“ZoneId”:1,

“StartDate”:”01-01-2020”,

“EndDate”:”31-12-2020”,

“LuminanceLevel”:80

},

{

“ZoneId”:1,

“StartDate”:”01-01-2021”,

“EndDate”:”31-12-2021”,

“LuminanceLevel”:85

}

]

}

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatAction	Repeat action command per interval	Number	10	0 to 24 hours	1	Set to 0 to send action command once.
NumberOfProfile	Number of profiles to be set	Number	10	1 to 10	5	
ZoneId	Lighting zone to apply setting when event is occurred	Number	10	1 to 63 / 255	255	Set to 255 to select all lights under a gateway

StartDate	Start date of event	String	20	dd-MM-yyyy dd – Day MM – Month yyyy – Year	01-01-2020	
EndDate	End date of event	String	20	dd-MM-yyyy dd – Day MM – Month yyyy – Year	31-12-2020	
LuminanceLevel	Set luminance level when event is occurred	Number	10	50 to 100	80	Value is in percentage

4.1.2.3.10 Daylight Profile

Daylight profile is used to control lighting operation mode based on ambient daylight, only one profile is supported to control lights under a gateway. For example, it can be used to activate sensor-based lighting control during overcast day as long as any of the designated photocell sensors indicate dim daylight.

Event Type	Command Type	Fields	Remarks
SmartLighting/setDaylightProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatAction, ZoneId, StartTime, EndTime, PhotoSensor1, PhotoSensor2, PhotoSensor3, DimThreshold, BrightThreshold, DimAction, BrightAction	Configure daylight profile (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatAction, ZoneId, StartTime, EndTime, PhotoSensor1, PhotoSensor2, PhotoSensor3, DimThreshold, BrightThreshold, DimAction, BrightAction	
SmartLighting/getDaylightProfile	Get		Get daylight profile settings (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatAction, ZoneId, StartTime, EndTime, PhotoSensor1, PhotoSensor2,	

		PhotoSensor3, DimThreshold, BrightThreshold, DimAction, BrightAction	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatAction	Repeat action command per interval	Number	10	0 to 250 min	30	Set to 0 to send action command once.
ZoneId	Lighting zone to apply setting when event is occurred	Number	10	1 to 63 / 255	255	Set to 255 to select all lights under a gateway
StartTime	Start time of event	String	20	HH:mm HH – Hour mm – Minute	07:00	
EndTime	End time of event	String	20	HH:mm HH – Hour mm – Minute	19:00	
PhotoSensor	Photo sensor id	String	255		SmartLighting-ST-600230-01010215	PhotoSensor2 and PhotoSensor3 are optional, set to “Nil” if not required
DimThreshold	Threshold to detect dim daylight	Number	10	50 to 1000	200	Value is in lux
BrightThreshold	Threshold to detect bright daylight	Number	10	50 to 1000	400	Value is in lux
DimAction	Perform lighting control under dim daylight	String	20	Auto / Force_On / Force_Off	Auto	
BrightAction	Perform lighting control under bright daylight	String	20	Auto / Force_On / Force_Off	Force_Off	

4.1.2.3.11 Rain Profile

Rain profile is used to control lighting operation mode based on rainfall, only one profile is supported to control lights under a gateway. For example, it can be used to keep lights fully on in surface car park as long as any of the designated rain sensors indicate heavy downpour.

Event Type	Command Type	Fields	Remarks
SmartLighting/setRainProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatAction,	Configure rain profile (unicast command to gateway)

		ZoneId, StartTime, EndTime, RainSensor1, RainSensor2, RainSensor3, LightRainThreshold, HeavyRainThreshold, LightRainAction, HeavyRainAction	
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatAction, ZoneId, StartTime, EndTime, RainSensor1, RainSensor2, RainSensor3, LightRainThreshold, HeavyRainThreshold, LightRainAction, HeavyRainAction	
SmartLighting/getRainProfile	Get		Get rain profile settings (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatAction, ZoneId, StartTime, EndTime, RainSensor1, RainSensor2, RainSensor3, LightRainThreshold, HeavyRainThreshold, LightRainAction, HeavyRainAction	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatAction	Repeat action command per interval	Number	10	0 to 250 min	30	Set to 0 to send action command once.
ZoneId	Lighting zone to apply setting when event is occurred	Number	10	1 to 63 / 255	255	Set to 255 to select all lights under a gateway
StartTime	Start time of event	String	10	HH:mm HH – Hour mm – Minute	19:00	
EndTime	End time of event	String	10	HH:mm HH – Hour mm – Minute	23:59	
RainSensor	Rain sensor id	String	255		SmartLighting-ST-600230-01010215	RainSensor2 and RainSensor3 are optional, set to “Nil” if not required
LightRainThreshold	Threshold to detect light rain	Number	10	5 to 255	5	Value is in terms of number of pulses per

						minute (each pulse is corresponding to 0.01mm of rainfall)
HeavyRainThreshold	Threshold to detect heavy rain	Number	10	5 to 255	30	Value is in terms of number of pulses per minute (each pulse is corresponding to 0.01mm of rainfall)
LightRainAction	Perform lighting control under light rain	String	20	Auto / Force_On / Force_Off	Auto	
HeavyRainAction	Perform lighting control under heavy rain	String	20	Auto / Force_On / Force_Off	Force_On	

4.1.2.3.12 Daylight Harvesting Profile

Daylight harvesting profile is used to adjust luminance level of each group of lights according to the ambient daylight measured using photocell sensor(s), only one profile is supported to control multiple daylight harvesting zones under a gateway. For example, it can be used to adjust luminance level of the lights near the entrance / exit of a basement car park.

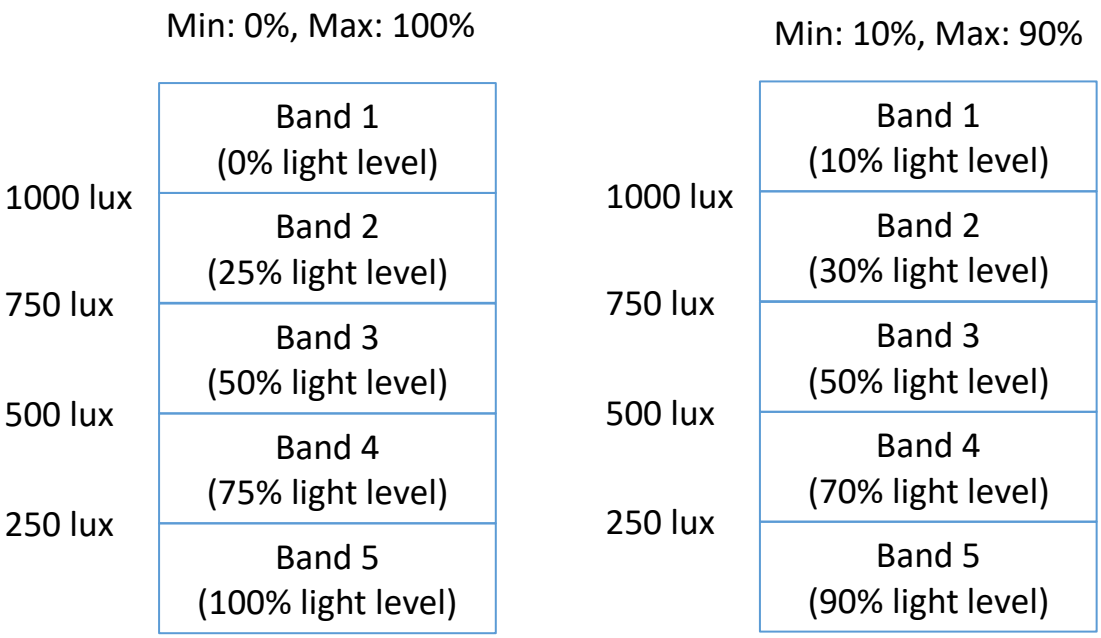
Event Type	Command Type	Fields	Remarks
SmartLighting/setDaylightHarvestingProfile	Set	Transmission, ConfigurationZone, ProfileControl, RepeatActivation, OverallZone, StartTime, EndTime, DayOfWeek, DaylightThreshold, MinLuminanceLevel, MaxLuminanceLevel	Configure daylight harvesting profile (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, ProfileControl, RepeatActivation, OverallZone, StartTime, EndTime, DayOfWeek, DaylightThreshold, MinLuminanceLevel, MaxLuminanceLevel	
SmartLighting/getDaylightHarvestingProfile	Get		Get daylight harvesting profile settings (unicast command to gateway)
	Get_Ack	ProfileControl, RepeatActivation, OverallZone, StartTime, EndTime, DayOfWeek, DaylightThreshold,	

		MinLuminanceLevel, MaxLuminanceLevel	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
ProfileControl	Profile control	String	20	Enable / Disable	Enable	
RepeatActivation,	Repeat activation command per interval	Number	10	0 to 250 min	30	Set to 0 to send activation command once.
OverallZone	Overall daylight harvesting zone which consists of the sensor nodes in all the daylight harvesting zones	Number	10	1 to 63	60	
StartTime	Start time of event	String	20	HH:mm HH – Hour mm – Minute	07:00	
EndTime	End time of event	String	20	HH:mm HH – Hour mm – Minute	19:00	
DayOfWeek	Select day of week for event to take effect	String	20	Weekday / Weekend / Everyday / Holiday	Everyday	
DaylightThreshold	Daylight threshold which is used to generate multiple daylight levels	Number	10	400 to 1000	1000	Value is in lux
MaxLuminanceLevel	Maximum luminance level corresponds to dim daylight	Number	10	50 to 100	90	Value is in percentage
MinLuminanceLevel	Minimum luminance level corresponds to bright daylight	Number	10	0 to 50	10	Value is in percentage

Note: The daylight threshold and min/max luminance level are used to generate 5 light bands for luminance adjustment based on measured ambient daylight. The daylight threshold is divided by 4 to generate 4 daylight levels whereas the difference of minimum and maximum luminance is divided by 4 to generate 4 luminance levels. For example:



4.1.2.3.13 Normal Light Automatics Test

The following commands are used to test normal light to detect light dim and flicker.

Event Type	Command Type	Fields	Remarks
SmartLighting/activateLightTest	Set	Transmission, ConfigurationZone	Activate lighting test. Alarm message will send out if light test failed.
	Set_Ack	Transmission, ConfigurationZone	
SmartLighting/setLightTestSchedule	Set	Transmission, ConfigurationZone, TestControl, StartTime	Setup a test schedule to test all lights under a gateway. Alarm message will send out if light test failed. (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, TestControl, StartTime	
SmartLighting/getLightTestSchedule	Get		Get schedule of lighting test. (unicast command to gateway)
	Get_Ack	TestControl, StartTime	
SmartLighting/getLightStatus	Get		Get light health status
	Get_Ack	LightStatus	

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
TestControl	Auto test control	String	20	Enable / Disable	Enable	
StartTime	Start time to perform lighting test	String	20	HH:mm HH – Hour Mm – Minute	03:00	
LightStatus	Health status of light	String	20	Normal / Dim / Flicker / Dim_Flicker / Off	Normal	

4.1.2.3.14 Emergency Light Automatics Test

The following commands are used to perform manual and auto test for emergency lights.

Event Type	Command Type	Fields	Remarks
SmartLighting/startELightTest	Set	Transmission, ConfigurationZone, TestType	Start emergency lighting test
	Set_Ack	Transmission, ConfigurationZone, TestType	
SmartLighting/stopELightTest	Set	Transmission, ConfigurationZone	Stop emergency lighting test
	Set_Ack	Transmission, ConfigurationZone	
SmartLighting/setELightAutoTest	Set	Transmission, ConfigurationZone, TestControl	Control auto test (unicast command to gateway)
	Set_Ack	Transmission, ConfigurationZone, TestControl	
SmartLighting/getELightAutoTest	Get		Get auto test control status (unicast command to gateway)
	Get_Ack	TestControl	
SmartLighting/setELightTestSchedule	Set	Transmission, ConfigurationZone, NumberOfProfile, [Profile 1] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime	Setup test schedules for auto test (unicast command to gateway)

		… [Profile n] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime	
	Set_Ack	Transmission, ConfigurationZone, NumberOfProfile, [Profile 1] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime … [Profile n] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime	
SmartLighting/getELightTestSchedule	Get		Get test schedules for auto test (unicast command to gateway)
	Get_Ack	NumberOfProfile, [Profile 1] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime … [Profile n] ProfileControl, ZoneId, FunctionalTestDate, FunctionalTestTime, DurationTestDate, DurationTestTime	
SmartLighting/getELightTestResult	Get		Get result of emergency lighting test
	Get_Ack	Test Time, TestType, TestResult, FailureType, NotTested, BatteryVoltage	

Note: [Profile #] in the above table is not the actual field, it is only meant for illustration purpose.

Sample parameters for SET command (SmartLighting/setELightTestSchedule)

```
"Parameters": {
    “CommandType”:”Set”,
    “Transmission”:”Unicast”,
    “ConfigurationZone”:0,
    “NumberOfProfile”:2,
    “AutoTestProfiles”: [
        {
            “ProfileControl”:”Enable”,
            “ZoneId”:1,
            “FunctionalTestDate”:”01-01-2020”,
            “FunctionalTestTime”:”01:00”,
            “DurationTestDate”:”01-15-2020”,
            “DurationTestTime”:”03:00”
        },
        {
            “ProfileControl”:”Enable”,
            “ZoneId”:2,
            “FunctionalTestDate”:”02-01-2020”,
            “FunctionalTestTime”:”01:00”,
            “DurationTestDate”:”02-15-2020”,
            “DurationTestTime”:”03:00”
        }
    ]
}
```

Sample parameters for GET command response (SmartLighting/getELightTestSchedule)

```
"Parameters": {
```

“CommandType”:”Get_Ack”,

“NumberOfProfile”:2,

“AutoTestProfiles”: [

{

“ProfileControl”:”Enable”,

“ZoneId”:1,

“FunctionalTestDate”:”01-01-2020”,

“FunctionalTestTime”:”01:00”,

“DurationTestDate”:”01-15-2020”,

“DurationTestTime”:”03:00”

},

{

“ProfileControl”:”Enable”,

“ZoneId”:2,

“FunctionalTestDate”:”02-01-2020”,

“FunctionalTestTime”:”01:00”,

“DurationTestDate”:”02-15-2020”,

“DurationTestTime”:”03:00”

}

]

}

Data Dictionary

Field Name	Description	Data Type	Max Length	Possible Values	Sample Values	Remarks
NumberOfProfile	Number of profiles to be set	Number	10	1 to 20	5	
ProfileControl	Test schedule control	String	20	Enable / Disable	Enable	
ZoneId	Lighting zone for auto test	Number	10	1 to 63	1	
FunctionalTestDate	Date of functional test	String	20	dd-MM-yyyy dd – Day MM – Month yyyy – Year	01-10-2020	
FunctionalTestTime	Time of functional test	String	20	HH:mm HH – Hour Mm – Minute	01:00	

DurationTestDate	Date of duration test	String	20	dd-MM-yyyy dd – Day MM – Month yyyy – Year	15-10-2020	
DurationTestTime	Time of duration test	String	20	HH:mm HH – Hour Mm – Minute	03:00	
TestTime	Date and time of lighting test	String	25	yyyy-MM-dd HH:mm:ss.SSS yyyy – Year MM – Month dd – Day HH – Hour 24hr format mm – Minute ss – Second SSS – Millisecond	2020-01-12 10:49:50.272	
TestType	Test type	String	25	Functional / Duration / Commissioning	Functional	
TestControl	Auto test control	String	20	Enable / Disable	Enable	
TestResult	Test result	String	25	Passed / Failed / Not_Tested / Test_In_Progress / Result_Not_Available	Passed	
FailureType	Type of failure	String	25	Light_Fail / Battery_Fail / Charger_Fail / Light_Dim / Light_Flicker / Light_Dim_Flicker	Battery_Fail	For test failed case
NotTested	Reason for not tested	String	50	Not_Responding / Emergency_Mode / Test_Button_Pressed / Light_Power_Not_Calibrated / Battery_Below_Nominal_Voltage / Battery_Not_Fully_Charged / Battery_Removed / Switch_Over_Fail / Incorrect_Nominal_Battery_Voltage / Battery_Low / Charger_fault	Battery_Removed	For not tested case
BatteryVoltage	Battery voltage after lighting test	Number	10	3.6 to 12	3.5	Value is in voltage

5 POINTS TO BE NOTED

- SensorId should be unique, as elaborated in Appendix III.
- If the sensor does not follow the ICD specifications, the Smart Hub will not show sensor readings or alarms.
- Prior to send sensor data, sensorIds must be added into master data and loaded in the Smart Hub system.
- Sensors must filter out the duplicate records in the flat files and empty readings.
- Records in a file must be sequential with respect to the time.

6 MASTER DATA CONFIGURATION

6.1 The Smart Hub system requires the master device list and basic properties of the each of the devices to be configured and linked to each device object. This is a one-off action. The data is populated in accordance to the physical device hierarchical relation.

The system is designed to take in large numbers of sensors properties without the upfront definition of individual sensor. As per the design, the properties of specific sensor can be represented as key value pair in static and dynamic parameters column of the master data. Hence any properties can be added to static properties.

The inputs in this excel will fall under:

1) Smart Hub Attributes – mandatory attributes required for Smart Hub integration

2) Sensor Attributes – attributes specific to the sensor (e.g. IOT sensor specifications, model no. , brand of sensor)

3) Specific Attributes – any other additional miscellaneous details associated with the sensor.

The Master Data establishes an identity for sensor/device and tracks metadata such as device's attributes, capabilities, geo info, resource path, and so on. This consistently formatted unique id would be a primary key which can be used as reference in event data or any data set.

6.2 Sensor Details for Smart Hub

The format below is an example. A detailed spreadsheet will be provided to the sensor vendor to populate.

Table 6-1: Master Data File definition

Smart Hub Attributes:

Property	Description	Sample Value
SensorId	<SensorType>-<Vendor>-<PostalCode>-<GenericIdentifier> Vendor name must be alphanumeric without special characters; Instead of Postal code, location code can be used if there is no postal code available for the location. E.g. JE-OUTDOOR A generic Identifier is used so that even when the specific sensor is replaced by new hardware, Smart Hub recognises it as the same sensor point.	See Appendix III
Sensor Type	Sensor type	Smart Lighting
Sensor Display Name	Name to be displayed in Smart Hub's real time monitoring software GUI	Appendix III
Postal Code	Postal code, if the postcode is not available just add some unique code eg: 99990001	600321
X	x/Easting	34.345234
Y	y/Northing	64.345454
Z	z/height	22.22
Location Descriptor	To display the location name in the smart hub system.	See Appendix III
Sensor Manufacturing Company	Brand/Vendor	ST Engineering
Sensor Model	Model	MHW33

Sensor Attributes:

Attributes specific to sensors will be required according to the sensor type. They are required for Smart Hub as well as for compliance with WOG sensor attribute conventions.

The sample values in the parameters/static properties column are for illustration purpose and they may not be real values of the sensors. The purpose of the static properties list to add static properties like Software Version, and compass bearing, etc.

In the master data, the attribute names (headers) should include the units.

For e.g. grid_frequency (Hz).

Specific Attributes:

These miscellaneous attributes are meant for any additional details specific to the sensor type or site-specific information such as the Mechanical & Electrical system details the sensor is tagged to.

Non-exhaustive examples are:

- Water pump motor specifications and installation date for Smart Pumps;
- Smart Lighting device location, gateway device no., connected light devices, light model;
- Water Meter sensor flowrate detection range, meter class, battery life etc.

6.3 Master Data Considerations

1. X, Y and Z coordinates shall be provided in svy21 reference system.
2. Any specific parameter with respect to sensor type can be added as key value pair, if you don't have any value at the moment just put it as N.A (Not applicable)

Appendix I – Glossary of Terms

GUI	Graphical User Interface
HDB	Housing & Development Board
ICD	Interface Control Document
JSON	JavaScript Object Notation
MQTT	Message Queue Telemetry Transport
Qos	Quality of Service
SMS	Short Messaging Service
SSL	Secure Sockets Layer
TLS	Transport Layer Security
URL	Uniform Resource Locator
WOG	Whole of Government

Appendix II– Masterdata Attributes Sample

This is a sample of the expected Masterdata attributes. The actual Masterdata document and its attributes will be in an excel document in table format.

The Contractor shall obtain the latest Masterdata document from HDB Building and Research Institute (BRI) and adhere to the latest format.



Master Data 20210801 Template.:

Water Meter Attributes (example)

	Attribute Name	Description	Sample Value
Smart Hub Attribute	**Sensor Id**	The sensor ID that serves as the unique identifier for each sensor Syntax: **SensorType-Vendor-Postalcode-GenericIdentifier** e.g. Water-Minerva-700123-WashBay1-Meter1	Water-minerva-600209-209-JEST21-Water-Wbay-01
	Sensor Type	To provide information on the sensor type or modality. For example, camera, noise, temperature, humidity, wind…etc.	Water Sub-Meter
	Sensor Display Name	To display the name in the smart hub system Syntax: **DisplayName** e.g. WaterMeter-WashBay1 (no spaces, use hyphen)	209JEST21
	Postal Code	If postal code is not available, location code should be given and this location code should be map to Town Council and to address	600209
	X Coordinate	To provide x-coordinate position of the sensor based on SVY21 reference system (minimum 3 decimal places).	17625.0213
	Y Coordinate	To provide y-coordinate position of the sensor based on SVY21 reference system (minimum 3 decimal places).	35555.6563
	Z Coordinate	To provide z-coordinate / height position (in metres) of the sensor based on SVY21 reference system (minimum 3 decimal places).	0
	Location Descriptor	To display the location name in the smart hub system Syntax : **Postalcode_LocationName** e.g. 700123_WashBay1 (no spaces, use underscore)	600209-Water-WBay-01
	Sensor Manufacturer Company	To provide manufacturer of the sensor	Sensus Advantech
	Sensor Model	To provide the model of the sensor	HRI & ADAM-6015

Sensor Specific Attribute	**Sensor Serial Number**	To provide the sensor's manufacturing serial number	Refer to SensorID
	Latitude	To provide latitude if unable to obtain X-coordinate based on SVY21 reference system.	
	Longtitude	To provide latitude if unable to obtain Y-coordinate based on SVY21 reference system.	
	Sensor Maintenance Company	To provide name of the contractor who is responsible for the maintenance	-
	Type of Deployment	Fixed or Mobile* [*Note: Mobile deployment means the sensor is portable and has no fixed location (e.g. Bodycam)]*	Fixed
	Existing_Planned_EnvisionedSensors	To indicate if the sensors are (a) Existing, (b) Planned (to be deployed in the next 5 years), or (c) Envisioned (in POC or research stage)	Existing
	Unit*	To provide the unit of measurement of the data collected by the sensor. For example, dB, °C, °F…etc. This field is required to be duplicated for each type of reading	Litres
	Range*	To provide the operational range at which the sensor is sensing or collecting the data e.g. 10-100 deg. This field is required to be duplicated for each type of reading unless it has been indicated otherwise	The sensors (i.e. HRI pulsars are attached directly on top of the surface of the water meters to collect the data)
	Sensor Coverage	To provide a brief estimation on the area the sensor is sensing or collecting data. For example 15m radius.	NA
	Power Consumption	To provide information on power consumed by the sensor (e.g. LR44 battery, 40W)	2W
	Power Source	To provide information on the power source. For example, solar, AC mains, DC mains, UPS/ Battery, POE…etc.	24DVC
	Network Connectivity	To provide information on network connectivity type. (e.g. 4G 3GB Plan, CAT5, Fibre, WiFi, Wireless Sensor Network)	CAT5 from sensor (i.e. ADAM) to ethernet port at switchroom
	Network Connectivity Provider	To provide information on the service provider for connectivity (e.g. StarHub, SingTel, M1, SPTel)	CAT5 from sensor (i.e. ADAM Controller) to ethernet port at switchroom
	Connectivity Contract Expiry	When the existing contract for network connectivity expires	Dependent on the service contract expiry of the electrical sub-meters
	Data Collection Mode	To indicate mode of sensor data collection: (a) Continuously in Real-Time, or (b) Every X minutes/hours, or (c) By Trigger	Every 1 min
	Data Transmission Mode	To indicate if the sensor is transmitting data: (a) Continuously in Real-Time, or (b) Every X minutes/hours, or (c) By Trigger	Every 15 mins

	Data Transmission	To indicate (a) Real-Time [RT] or (b) Historical [H], for data transmission requirement.	Historical (1 file containing 15 lines of data captured every minute will be sent every 15 mins).
	Bandwidth	Bandwidth required by sensor for data transmission (e.g. 10 Mbps, 10Kbps)	
	Data Usage Per Month (Mbytes)	Total data usage (sent and received) by each sensor per month (e.g. 0.1 MB, 100 MB)	3.1 MB (based on estimate of 100Kb per day)
	Availability (%)	Service Level Agreement (SLA) - Network Availability (e.g. ≥ 99%)	
	Compass Bearing	To provide indicative direction (in degrees based on True North) where the sensor is looking	Not applicable
	Resiliency	(E.g. (a) Path diversity, (b) Exchange Diversity, (c) Path & Exchange Diversity, (d). Nil)	
	Security	Security measures implemented for network traffic, if any (E.g. VPN, VPN + Encryption, Nil)	
	Installation Date of Sensor Deployment	To provide installation date (dd/mm/yy) of sensor deployment.	to be installed
	Maintenance Frequency	To provide the maintenance frequency of the sensor device (i.e. 1 Year, 2 Years etc…)	Every 5 years
	Quantity Per Location	Not applicable for Mobile sensors. To fill in if there are 2 or more sensors in the same location	1 per location
Miscellaneous Attribute	**Maximum Flow Rate**	To indicate the max range in which the water sub-meter is able to read	3 L/hr
	Minimum Flow Rate	To indicate the min range in which the water sub-meter is able to read	3125 L/hr
	Water Tariff	The Tariff imposed on water consumption	
	Class Type	To indicate the class type of the water-submeter	Class C
	Meter Type	To indicate the meter type of the water-submeter	Volumetric
	Service Type	To indicate whether the tap that the water submeter is monitoring is serving residential or common services	Common services
	Pulsar Type	Type of pulsar	Inductive
	Resolution	To indicate the Pulse Weight specification	1Litre/pulse
	Battery Life	To indicate the battery life of the Pulsar	approx. 5-10 years

Appendix III – Nomenclature for Sensor Naming and Masterdata

Smart System	**SenderId** (use hyphens)	**SensorId** (use hyphens)	**Sensor Display Name** (no space, use hyphen only if there is hierarchy)	**Location Descriptor** (no space, use underscore between postal code and location name)
Smart Water Pump	**SensorType-Postalcode-AggregatorIdentifier** e.g. Pump-700123-BoosterPumpRoom	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. Pump-Ademco-700123-BoosterPumpRoom-ControlPanel Pump-Ademco-700123-BoosterPumpRoom-Pump1	**DisplayName** e.g. ControlPanel Pump1	**Postalcode_LocationName** e.g. 700123_BoosterPumpRoom
Smart Water Sub-Meter	**SensorType-Postalcode-AggregatorIdentifier** e.g. Water-700123-WashBay1	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. Water-Minerva-700123-WashBay1-Meter1	**DisplayName** e.g. WaterMeter1	**Postalcode_LocationName** e.g. 700123_WashBay1
Power Consumption Fault Detection Monitoring System (PCFDMS)	**SensorType-Postalcode-AggregatorIdentifier** e.g. PCFDMS-700123-1 [1 or 2 for the different Consumer Switch Room]	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. PCDFMS-Gathergates-700123-1-Q1-M1 PCDFMS-Gathergates-700123-1-Q1-ESB1 (At individual sensor level)	**DisplayName** e.g. PCFDMS-Q1-M1	**Postalcode_LocationName** e.g. 700123_SwitchRoom1

Smart Lighting System	**SensorType-Postalcode-AggregatorIdentifier** e.g. SmartLighting-700123-01	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. SmartLighting-ST-700123-01010201 SmartLighting-ST-700123-01010202 SmartLighting-ST-700123-01010203	**DisplayName** e.g. MotionSensor-1 MotionSensor-2	**Postalcode_LocationName** e.g. 700123_LiftLobby 700123_Level1
Urbanwater Harvesting System	**SensorType-Postalcode-AggregatorIdentifier** e.g UWHS-700123-RainwaterTreatmentRoom	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. UWHS-ACE-700123-PUBTopup UWHS-ACE-700123-Washing UWHS-ACE-700123-Irrigation	**DisplayName** e.g. WaterMeter-PUBTopup WaterMeter-Washing WaterMeter-Irrigation	**Postalcode_LocationName** e.g. 700123_RainwaterTreatmentRoom
Smart Irrigation	**SensorType-Postalcode-AggregatorIdentifier** e.g. SmartIrrigation-700123-B1	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. SmartIrrigation-ATS-700123-B1-1 SmartIrrigation-ATS-700123-B1-2 SmartIrrigation-ATS-700123-L2-1	**DisplayName** e.g. Moisturesensor-1 Moisturesensor-2	**Postalcode_LocationName** e.g. 700123_B1 700123_L2

Smart Rainwater Sump Pump	**SensorType-Postalcode-AggregatorIdentifier** e.g. RWSPump-700123-B1	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. RWSPump-Company-700123-B1-1	**DisplayName** e.g. RainwaterSumpPump-1	**Postalcode_LocationName** e.g. 700123_Basement1
Carpark Mechanical Ventilation Monitoring	**SensorType-Postalcode-AggregatorIdentifier** e.g. CarparkMV-700123-B1	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. CarparkMV-Colt-700123-B1-1	**DisplayName** e.g. CarparkMV-COsensor-1	**Postalcode_LocationName** e.g. 700123_CarparkLevelB1
Fast and Secure Access System (Smart Lock)	**SensorType-Postalcode-AggregatorIdentifier** Same as SensorId	**SensorType-Vendor-Postalcode-GenericIdentifier** e.g. SmartLock-CA-700123-SwitchRoom-Lock SmartLock-CA-700123-PumpRoom-Lock	**DisplayName** e.g. SmartLock-PumpRoom	**Postalcode_LocationName** e.g. 700123_SwitchRoom, 700123_PumpRoom, 700123_RoofTop